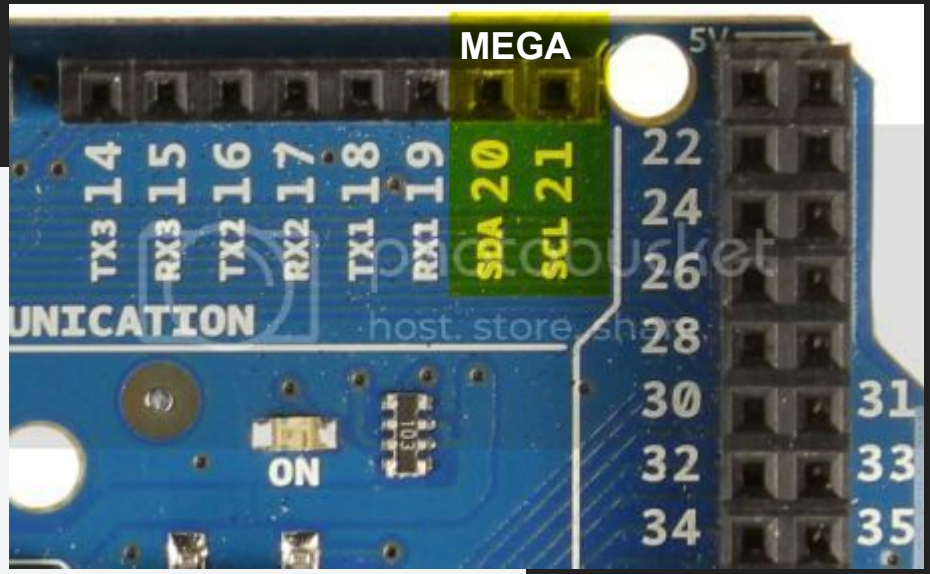
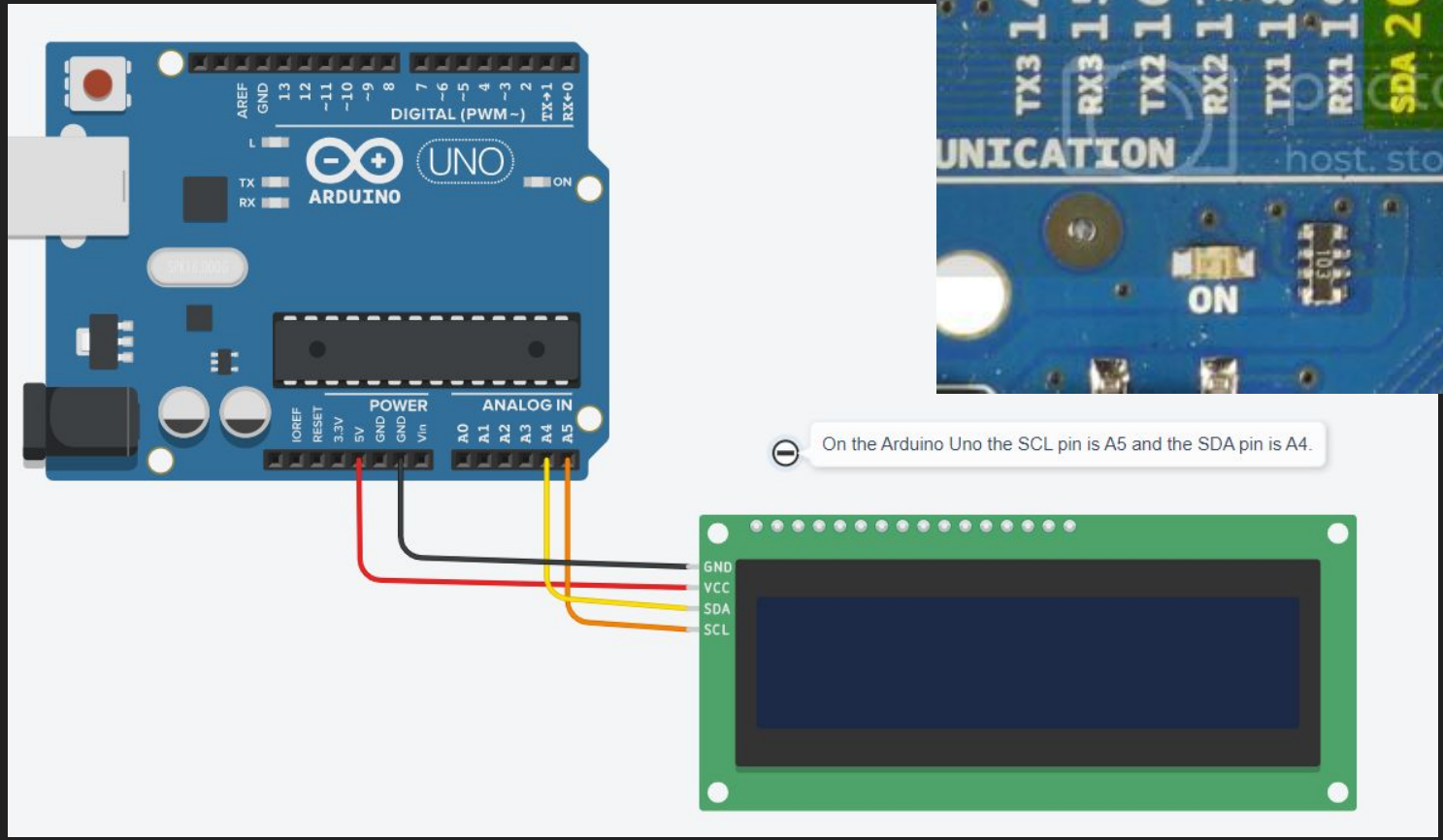
Digital Electronics Interfacing

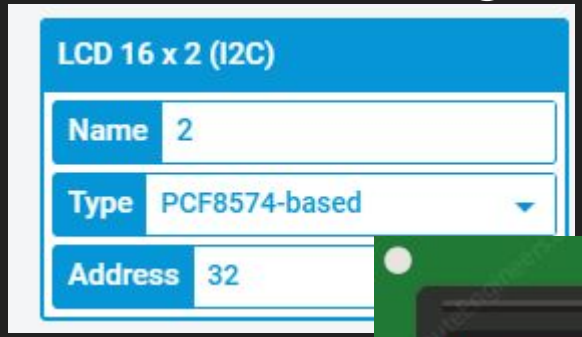
Lesson 9

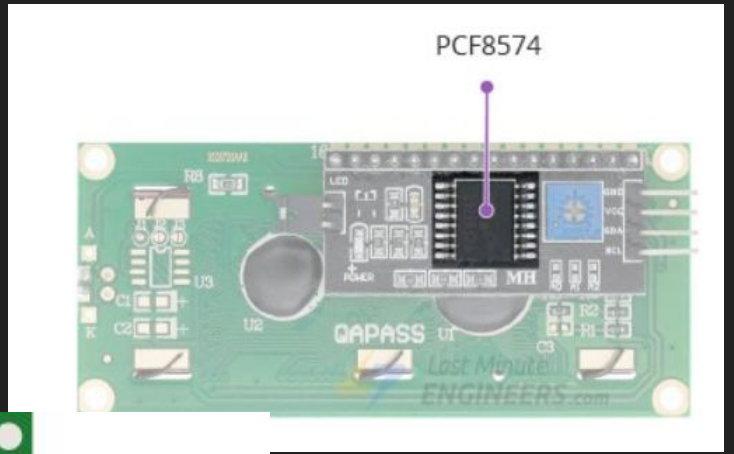
I2C LCD-Circuit

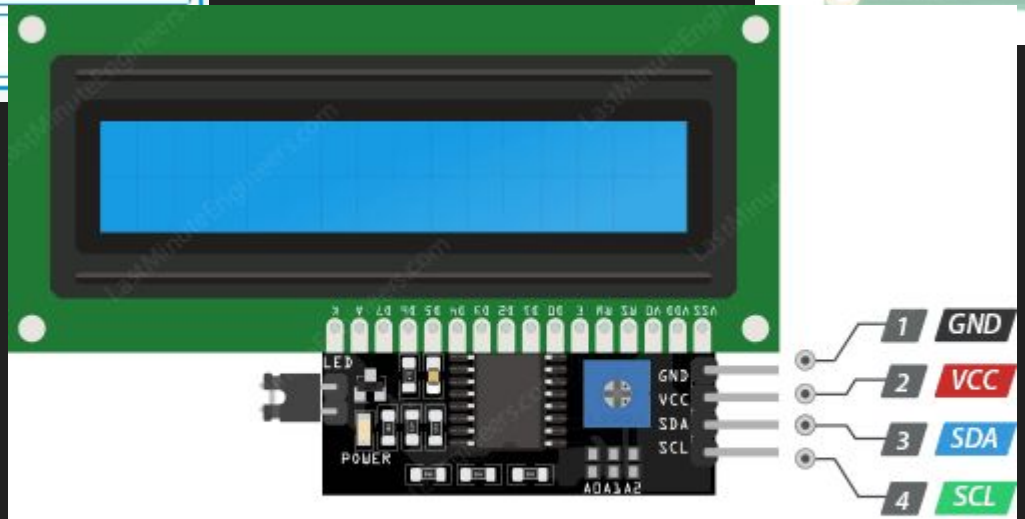




I2C LCD-Settings







I2C LCD Pinout

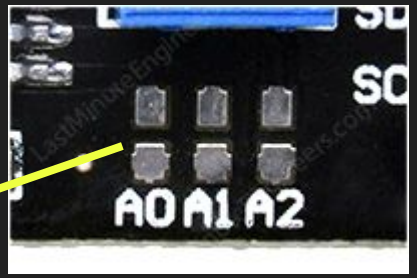
Last Minute ENGINEERS.com

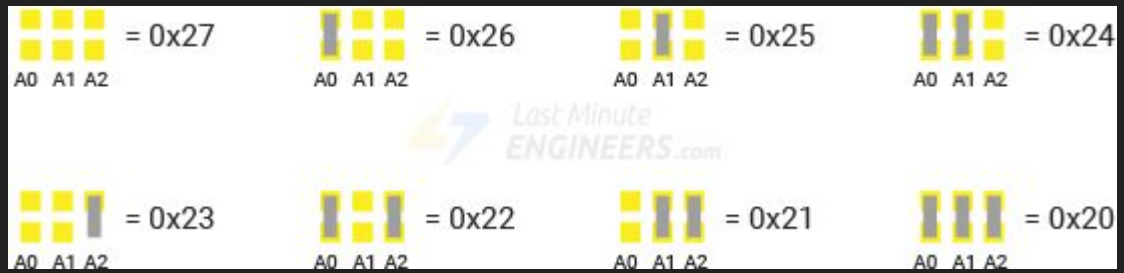
I2C Device Addresses

Scanner Code-will detect and display the hex I2C addresses of all connected I2C devices (most of our I2C LCDs in the class have the address **0x27**)

https://playground.arduino.cc/Main/I2cScanne

Setting the I2C Address of the LCD





https://lastminuteengineers.com/i2c-lcd-arduino-tutorial/

I2C LCD-Code

```
#include <Wire.h>
/*The sketch begins by including the LiquidCrystal_I2C library.*/
#include <LiquidCrystal_I2C.h>
/*LiquidCrystal_I2C(address, columns, rows)*/
//32 is the hex address...need to find this
//your LCD probably has I2C address 0x27Hex or 0x3FHex
//To find the address here just select the LCD and see the selected
//Also, you need to make sure what type is selected.
//In this case only the PCF8574 type is being used.
LiquidCrystal_I2C lcd(32,16,2);



void setup()
{
  /*
  In 'setup' we call three functions.
  The first function is init().
  It initializes the LCD object.
  The second function is clear().
  This clears the LCD screen and moves the cursor to the top left c
  And third, the backlight() function turns on the LCD backlight.
  */
  lcd.init();
  lcd.clear();
  lcd.backlight();
  //set the cursor position to the third column of the first row by
  lcd.setCursor(2,0);
  lcd.print("Hello world 0!");

}

void loop()
{
  delay(1000);
  lcd.clear();
  //set the cursor position to the third column of the first row by
  lcd.setCursor(2,0);
  lcd.print("Hello world 1!");
  delay(1000);
  lcd.clear();
  //set the cursor position to the third column of the first row by
  lcd.setCursor(2,0);
  lcd.print("Hello world 2!");
}
```

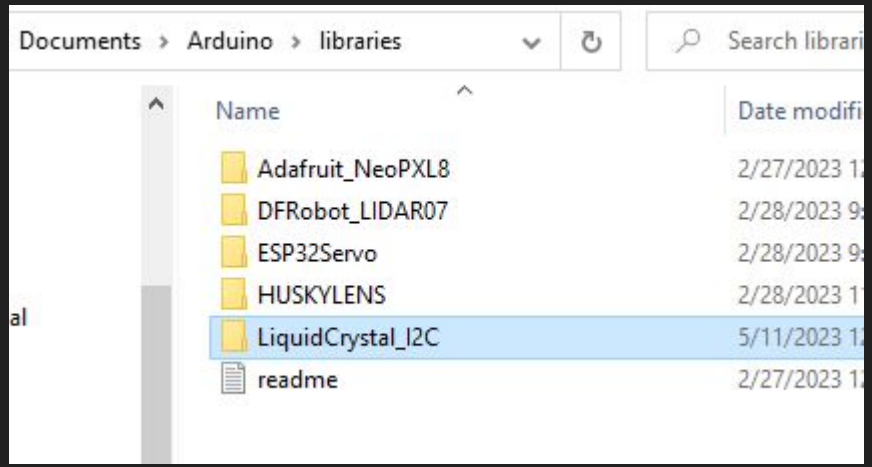
Arduino If Statements

```
1  if (someCondition) {
2  // do stuff if the condition is true
3  } else {
4  // do stuff if the condition is false
5  }
```

```
if (someCondition) {
// do stuff if the condition is true
} else if (anotherCondition) {
// do stuff only if the first condition is false
// and the second condition is true
}
```

Adding the Library (https://drive.google.com/file/d/1gNWgBpUb4xeCcE35kAdAbjSXvOu89xsm/view?usp=sharing)

To use external libraries you must first download them. They'll usually be in zip format. Extract them to the Documents/Arduino/Libraries folder. Restart the Arduino IDE and you should now be able to use the library in your code.



Assignment (create the tinkercad prototype and the actual circuit)
/15 Tinkercad Protoype /15 Physical Circuit - App

https://www.tinkercad.com/things/dEBx4FItg8X-powerful-wolt

Create a circuit with 2 LEDs and an I2C LCD. Complete the following circuit as shown in the video:

